Class Test 2	12 April 2013 FSK116	*Klastoets 2*

Total/Totaal: 35 max

Surname *MEMO* **Van**	**Initials** M **Voorletters** **Stud. No/nr.** 0000001
The undersigned hereby confirms that he/she is familiar with and accepts the examination regulations.	Ondergetekende bevestig hiermee dat hy/sy op hoogte is met die eksamenregulasies en dat hy/sy hom/haar daaraan onderwerp.

Handtekening ______________________________ *Signature* *Date*______________________ *Datum*

$v_{avg} = ½(v_0 + v)$ $v = v_0 + at$ $x = x_0 + v_0 t + ½a_x t^2$ $x = x_0 + v_{avg,x} t$	$v_x^{\ 2} = v_{ox}^2 + 2a_x(x - x_0)$ $x - x_0 = ½(v_{0x} + v_x)t$ $x - x_0 = v_x t - ½a_x t^2$	$a = \frac{v^2}{r}.$ $g = 9{,}8 \text{ m/s}^2$ $h = \frac{v_0^2 \sin^2\theta}{2g}$	$R = \frac{v_0^2}{g}\sin 2\theta_0$ $T = \frac{2v_0 \sin\theta}{g}$ $D = ½CA\rho v^2$

Question 1/*Vraag 1 (Q 11 Homework)* **[9]**

The position $\vec{r}$ of a particle moving in an xy plane is given by $\vec{r} = (2.00t^3 - 5.00t)\hat{\mathrm{i}} + (6.00 - 7.00t^4)\hat{\mathrm{j}}$ with $\vec{r}$ in meters and t in seconds. In unit vector notation, calculate the following at t = 2.00 s:

Die posisie van 'n deeltjie wat in die xy-vlak beweeg word gegee deur $\vec{r} = (2.00t^3 - 5.00t)\hat{\mathrm{i}} + (6.00 - 7.00t^4)\hat{\mathrm{j}}$, *waar* $\vec{r}$ *in meter en t in sekondes is. In eenheidsvektor-notasie, bereken die volgende by t =2.00 s:*

(a) $\vec{r}$. [2]

$\vec{r} = \left[(2.00t^3 - 5.00t)\hat{\mathrm{i}} + (6.00 - 7.00t^4)\hat{\mathrm{j}}\right]$ m At $t = 2.00$ s: $\vec{r} = \left[(2.00(2.00)^3 - 5.00(2.00))\hat{\mathrm{i}} + (6.00 - 7.00(2.00)^4)\hat{\mathrm{j}}\right]$m $= \left[(6.00)\hat{\mathrm{i}} + (-106.00)\hat{\mathrm{j}}\right]$m	✓✓

(b) $\vec{v}$. [4]

$\vec{r} = \left[(2.00t^3 - 5.00t)\hat{\mathrm{i}} + (6.00 - 7.00t^4)\hat{\mathrm{j}}\right]$m $\vec{v} = \frac{d\vec{r}}{dt} = \left[(6.00t^2 - 5.00)\hat{\mathrm{i}} + (-28.00t^3)\hat{\mathrm{j}}\right]\text{ms}^{-1}$	✓✓
At $t = 2.00$ s: $\vec{v} = \left[(6.00(2.00)^2 - 5.00)\hat{\mathrm{i}} + (-28.00(2.00)^3)\hat{\mathrm{j}}\right]\text{ms}^{-1}$ $= \left[(19.0)\hat{\mathrm{i}} + (-224)\hat{\mathrm{j}}\right]\text{ms}^{-1}$	✓✓

(c) What is the angle between the positive direction of the x axis and the direction of the particle's direction of motion at t = 2.00 s?
Wat is die hoek tussen die positiewe x-as en die rigitng van die deeltjie se beweging by t = 2.00 s? [3]

$\vec{v}=\left[(19.0)\hat{\mathrm{i}}+(-228)\hat{\mathrm{j}}\right]\mathrm{ms}^{-1}$ $\text{Tan}\,\theta=\frac{y}{x}$ $\theta=\tan^{-1}\frac{-228}{19}=-12$ $\theta=-85°$	✓ ✓✓

Question 2 / *Vraag 2* (In class) **[6]**

A rifle with a muzzle velocity v_0 = 450 m/s shoots a bullet at a target 50 m away.
'n Koeël word afgevuur uit 'n geweerloop teen 'n spoed v_0 = 450 m/s. Die teiken is 50 m weg vanaf die geweerloop opening.

(a) Assume the muzzle and the target are at the same height, how high above the target must the barrel be pointed in order to hit the target?
Veronderstel dat die geweerloop en die teiken op dieselfde hoogte is, hoe hoog bo die teiken moet die geweer gerig word om die teiken te tref? [4]

Since the launch and landing points are on the same level, we can use the range equation to determine the launch angle (angle at which the bullet leaves the barrel. $R=\frac{v_0^2}{g}\sin 2\theta_0$ $2\theta_0=\sin^{-1}\left(\frac{Rg}{v_0^2}\right)=0.138°$ $\theta_0=0.07°$	✓ ✓ **Can use equations of motion to determine the v_{ox} *and* v_{oy} components and then determine the launch angle of v_o.**
The target is 50 m away, and therefore using the Tan function, the height above the target can be determined. $\tan\theta=\frac{h}{50}$ $h=60\text{ mm}$	✓ ✓

(b) Determine the maximum height reached by the projectile.
Bepaal die maksimum hoogte wat deur die projektiel bereik word. [2]

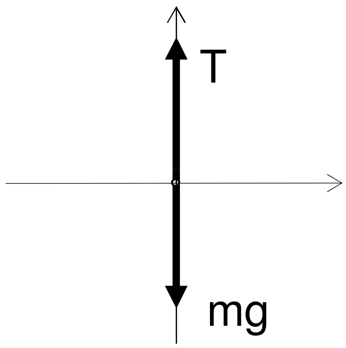
Since this is a projectile with the launch and landing heights on the same level, we can use the following equation: $H=\frac{v_0^2\sin^2\theta_0}{2g}=15\text{ mm}$	✓ ✓
Alternative method: $v_{0y}=v_0\sin\theta=450\sin(0.07°)=0.55\text{ m/s}$	✓

At the maximum height, the velocity in the y direction will be $v_y = 0$ m/s. $v_y{}^2 = v_{oy}^2 + 2a_y(y - y_0)$ $\Delta y = \frac{v_y{}^2 - v_{oy}^2}{2a_y} = \frac{(0\text{ m/s})^2 - (0.55\text{m/s})^2}{2(-9.8\text{m/s}^2)} = 0.015\text{ m}$	✓

Question 3 / *Vraag 3* **[7]**

An elevator cab and its load have a combined mass of 1600 kg. It is moving downward initially at 12 m/s, and is brought to rest with constant acceleration in a distance of 42 m. Determine the tension in the cable.

'n Hysbak met 'n vrag het 'n totale massa van 1600 kg. Dit beweeg aanvangklik afwaarts teen 12 m/s, en word tot stilstand gebring teen 'n konstante versnelling oor 'n afstand van 42 m. Bepaal die spankrag in die kabel. [2+5]

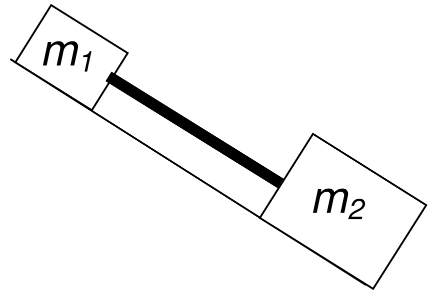


We can first determine the net acceleration – which is in the upward direction since the velocity is downward. v_i = -12 m/s Δy = -42 m v_f = 0 m/s $v^2 = v_0^2 + 2a\Delta y$ $a = \frac{v^2 - v_0^2}{2\Delta y}$ $= \frac{0 - (-12\text{ m/s})^2}{2(-42\text{ m})} = 1.71\text{ m/s}^2$	**✓ Use equations of motion to determine *a*** ✓
With the net acceleration (upwards and therefore positive in a conventional set of axes) we know the net force and can then apply Newton's second law: $\sum F_y = ma_y = T - mg$ $T = ma_y - mg$ $= m(a - g) = 1600\text{ kg}(-1.71\text{ m/s}^2 - 9.8\text{ m/s}^2)$ $= 1.84 \times 10^4\text{ N}$	✓ ✓ ✓

Question 4 / *Vraag 4* **[14]**

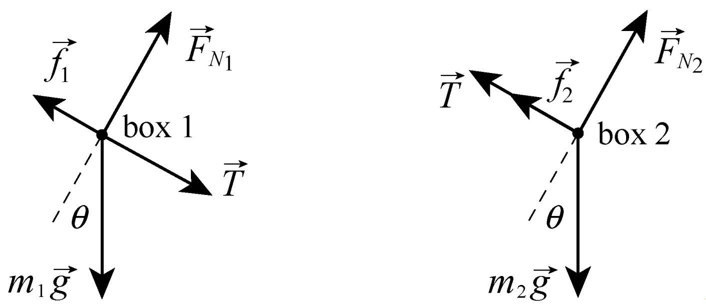
In the Figure, a box of Fizz Pops (mass m_1 = 1.65 kg) and a box of Aeros (mass m_2 = 3.30 kg) slide down an inclined plane while attached by a massless rod parallel to the plane. The angle of the plane is θ = 30.0 °. The coefficient of kinetic friction between the Fizz Pop box and the incline is μ_1 = 0.226; and that between the Aero box and the incline is μ_2 = 0.113.

Die Figuur dui 'n boks Fizz Pops (massa m_1 = 1.65 kg) en 'n boks Aeros (massa m_2 = 3.30 kg) aan wat teen 'n skuinsvlak afgly. Die twee bokse is aan mekaar gekoppel met 'n massalose staaf wat parallel met betrekking tot die vlak is. Die hoek waarteen die vlak gerig is, is θ = 30.0 °. Die kinetiese wrywingskoëffisient tussen die Fizz Pop boks en die skuinsvlak is μ_1 = 0.226; en dié tussen die Aero boks en die skuinsvlak is μ_2 = 0.113.



(c) Determine the tension in the rod and the magnitude of the acceleration of the two boxes.
Bepaal die spankrag in die staaf en die groote van die versnelling van die twee bokse.

[2+10]



✓✓

For the Fizz Pop Box

$$\sum F_y = m_F a_y = N_F - m_F g\cos\theta = 0$$

$$N_F = m_F g\cos\theta$$ ✓✓

$$\sum F_x = m_F a_x = T + m_F g\sin\theta - f_{kF}$$

$$a_x = \frac{T}{m_F} + g\sin\theta - \mu_{kF} g\cos\theta$$ ✓ ①

For the Aero Box

$$\sum F_y = m_A a_y = N_A - m_A g\cos\theta = 0$$

$$N_A = m_A g\cos\theta$$ ✓✓

$$\sum F_x = m_A a_x = m_A g\sin\theta - T - f_{kA}$$

$$m_A a_x = m_A g\sin\theta - T - \mu_{kA} N_A$$

$$a_x = g\sin\theta - \frac{T}{m_A} - \mu_{kA} g\cos\theta$$ ✓ ②

① = ②

$$a_x = \frac{T}{m_F} + g\sin\theta - \mu_{kF} g\cos\theta = g\sin\theta - \frac{T}{m_A} - \mu_{kA} g\cos\theta$$

$$\frac{T}{m_F} + - \frac{T}{m_A} = \mu_{kF} g\cos\theta - \mu_{kA} g\cos\theta$$

$$\left(\frac{1}{m_F} + - \frac{1}{m_A}\right)T = g\cos\theta\left(\mu_{kF} - \mu_{kA}\right)$$ ✓✓✓

$$T = \frac{g\cos\theta\left(\mu_{kF} - \mu_{kA}\right)}{\left(\frac{1}{m_F} + - \frac{1}{m_A}\right)}$$

$$= 1.05 \text{ N}$$

$$a_x = \frac{T}{m_F} + g\sin\theta - \mu_{kF}\, g\cos\theta = g\sin\theta - \frac{T}{m_A} - \mu_{kA}\, g\cos\theta$$

$$a_x = \frac{1.05\,\text{N}}{1.65\ \text{kg}} + g\sin 30° - 0.226g\cos 30° \quad ✓$$

$$= 3.62\,\text{m.s}^{-2}$$

(b) In words, describe how the tension and the acceleration would change if the boxes swopped positions (Areo box at the higher position).
In woorde, beskryf hoe die spankrag en die versnelling sou verander indien die bokse se posisies omruil (Aero boks by die hoër posisie).

[2]

According to the equations for T and a above, we just have to swop the labels (F and A).
This will give a negative T ✓ (Compression and not tension) and the acceleration will be the same as before✓

THE END / DIE EINDE